Fiscal Year:	FY 2016	**Task Last Updated:**	FY 04/07/2016
PI Name:	Gibble, Kurt Ph.D.		
Project Title:	Contributions to the Evaluation of the ACES Clock PHARAO		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	16802-6300	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	06/08/2012	**End Date:**	12/31/2018
No. of Post Docs:		**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2018 per PI (Ed., 3/10/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	ACES (Atomic Clock Ensemble in Space) is a European Space Agency (ESA) project that will test fundamental physics, search for the time variation of fundamental constants, and enable highly precise frequency comparisons between clocks on Earth. It is planned to operate for at least 18 months on the International Space Station. PHARAO (Projet d'Horloge Atomique par Refroidissement d'Atomes en Orbit) is the laser-cooled, microgravity, atomic cesium clock of ACES that provides state-of-the-art accuracy, with a goal of 1 part in 10^(16) accuracy. We propose to assist in the accuracy evaluation of PHARAO, a central scientific objective of the ESA mission. We will contribute to the analysis of the three largest sources of systematic uncertainty: clock frequency shifts due to the collisions of ultracold atoms, a Doppler shift known as the distributed cavity phase shift, and a frequency shift due to the lensing of the atomic wavepackets by the clock's microwave field. The current working uncertainty budget projects 2.2×10^(-16) accuracy. The proposed work		

	will reduce the largest contribution to the clock's uncertainty to negligible levels and investigate significant reductions of the others, with a goal of reaching 10^(-16) accuracy. The work will help define the limits of PHARAO's accuracy and identify workable procedures to demonstrate its accuracy.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work on microwave cavities produced an improved design that will be used for Cs and Rb fountain clocks. These are being incorporated into new clocks being built at the British, South Korean, and Canadian National Laboratories, National Physical Laboratory (NPL), KRISS (Korea Research Institute of Standards and Science) , and NRC. The new cavity design allows significantly reduced clock errors due to Doppler shifts. Our work has also led to a deeper understanding of the microwave lensing frequency shift that is being used as a correction of International Atomic Time and, more broadly, contributes to scientific progress towards anticipated future optical frequency atomic clocks.
Task Progress:	This year's effort directly related to PHARAO focused on evaluating the microwave lensing frequency shift and frequency shifts due to cavity phase variations. On microwave lensing, we finalized the calculations of microwave lensing, including optimizing integration routines using a Monte-Carlo technique as well as developing accurate analytic models with various levels of approximations. We have demonstrated a small sensitivity of the microwave lensing shift to the PHARAO parameters, the selection cavity inhomogeneities, and its perturbation due to low frequency vibrations of the International Space Station (ISS). Our work showed that the microwave lensing behaves significantly differently than this shift of fountain clocks on Earth. We obtain a fractional frequency shift of PHARAO of 11.4×10^-17 to 13×10^-17, slightly larger than the accuracy goal of PHARAO. Our conservative uncertainty of +/- 4×10^-17 for the microwave lensing contributes negligibly to the updated PHARAO frequency accuracy. Our paper, in collaboration with our French colleagues, was published in Metrologia. Related to the PHARAO microwave lensing frequency shift, we published an arXiv paper and a Reply to a Comment from the National Institute of Standards and Technology (NIST) regarding our microwave lensing treatment. The recent predictions of the NIST treatment differ significantly from ours. We have identified specific problems in their treatment and why they lead to misconceptions. Our treatment is used by all but a couple of the primary atomic clocks around the world that contribute to the international atomic timescale. A number of evaluations are in progress and all of these that we are aware of, including PHARAO, are using the corrections that we calculate. Significant progress has been made on evaluating the PHARAO frequency uncertainty due to cavity phase (CP) variations. We have performed numerous simulations of clock shifts due to CP, from both the known wall losses as well as potential inhomogeneities of the wall losses. As for microwave lensing, these shifts behave significantly differently than for fountain clocks and our preliminary results suggest that a new technique to evaluate them, which requires a microgravity environment, will be effective. Following our microwave lensing evaluation, CP is the remaining large uncertainty in the PHARAO accuracy budget. In collaboration with the French PHARAO team, we have compared our results to measurements and assisted with the integration of our precise microwave fields into their simulation software. Current work is focused on analyzing the instability of the clock versus key parameters so that we can optimize the CP accuracy evaluation. A manuscript is in preparation, in collaboration with our French colleagues. Related to the cold collision frequency shift of PHARAO, we have finalized precise measurements of Cesium scattering phase shifts in an atomic fountain. We have evaluated a subtle systematic error due to phase shifts of unscattered atoms and have worked on comparing our results to theoretical results. We generally see extremely good agreement, but our high precision elucidates some small differences. Work has analyzed the uncertainties in the theoretical results, as well as their implications for the PHARAO cold collision shift. Aaron Bennett defended his Ph.D. thesis on this work and publications drafts are being prepared for submission. More broadly, some effort has explored future microgravity atomic clock candidates based on optical frequencies that have small sensitivities to electric fields. Electric fields due to cosmic rays are more important in microgravity, as are those due to blackbody radiation at non-cryogenic temperatures. We have participated in initial research on magnesium and cadmium optical lattice clocks. Like mercury, these feature a much lower blackbody and stray electric field sensitivity than other candidates, including strontium and ytterbium.
Bibliography Type:	Description: (Last Updated: 05/28/2026)
Articles in Other Journals or Periodicals	Gibble K. "Comment on 'Frequency shifts in NIST Cs primary frequency standards due to transverse rf field gradients.' " Cornell University Library, May 4, 2015. arXiv:1505.00691 [physics.atom-ph]; http://arxiv.org/abs/1505.00691 , May-2015
Articles in Peer-reviewed Journals	Kulosa AP, Fim D, Zipfel KH, Rühmann S, Sauer S, Jha N, Gibble K, Ertmer W, Rasel EM, Safronova MS, Safronova UI, Porsev SG. "Towards a Mg lattice clock: Observation of the 1S0-3P0 transition and determination of the magic wavelength." Physical Review Letters. 2015 Dec 11;115(24):240801. PubMed PMID: 26705620 ; http://dx.doi.org/10.1103/PhysRevLett.115.240801 , Dec-2015
Articles in Peer-reviewed Journals	Gibble K. "Reply to `Comment on `Ramsey spectroscopy, matter-wave interferometry, and the microwave-lensing frequency shift.' " Physical Review A. 2015 Jun;91(6):067602. http://dx.doi.org/10.1103/PhysRevA.91.067602 , Jun-2015
Articles in Peer-reviewed Journals	Peterman P, Gibble K, Laurent Ph, S, Salomon Ch. "Microwave lensing frequency shift of the PHARAO laser-cooled microgravity atomic clock." Metrologia. 2016 Mar;53(2):899-907. http://dx.doi.org/10.1088/0026-1394/53/2/899 , Mar-2016
Articles in Peer-reviewed Journals	Gibble K. "Systematic effects in atomic fountain clocks." Journal of Physics: Conference Series. 2016;723:012002. (8th Symposium on Frequency Standards and Metrology 2015.) https://doi.org/10.1088/1742-6596/723/1/012002 [Reported previously in April 2016 as IOP Conference Series, in press as of April 2016.] , Jul-2016
Dissertations and Theses	Bennett A. (Aaron Bennett) "Precision Measurements of Quantum Scattering Phase Shifts and Their Variation Through Feshbach Resonances." Dissertation, The Pennsylvania State University, December 2015. , Dec-2015